

also more complex in comparison. In addition to the digital and analogue signal processing circuits fabricated using the complementary metal-oxide semiconductor process, a specialised TMR material layer is deposited on top of the silicon wafer. This additional step, which is not required for Hall sensors, necessitates tailored equipment and increases production costs. The TMR layer is critical for creating smaller, higher-impedance sensors, thereby improving their signal-to-noise ratio.

What technical challenges do integrated current sensors encounter?

Integrated current sensors generally work well but can be affected by electromagnetic interference (EMI). High-power applications need specialised designs to manage EMI. Achieving high accuracy and precision in current measurements is challenging due to temperature variations, noise, and offset errors. Measuring very high currents requires a bus bar, which poses its own challenges.

How do you deal with electrical noise?

Proper printed circuit board (PCB) design guidelines are provided to ensure optimised current measurement. When using bus bars, it is recommended to cut shapes such as S and U into the bus bars. This increases current density, resulting in stronger magnetic fields that can be measured more accurately. Sometimes, sensors are placed on either side of the notches to cancel common-mode noise and stray-field interference.

What testing methods do you use to ensure your product functions correctly before release?

We check our designs in multiple ways during the design phase to make sure they work as expected. There are simulations in the early design phases, followed

by verification and validation tests before it goes into manufacturing. Once the integrated circuit returns from manufacturing, bench and automated tests are conducted to validate all features and specifications.

How do you evaluate and improve the performance of your products over time?

Continuous improvement is essential as market demands evolve and customers use our products in new ways, presenting new challenges—such as increased sensitivity requirements. For instance, while a 1 per cent total measurement error was once acceptable, customers now expect 0.5 per cent accuracy. We conduct ongoing experiments to assess potential improvements and keep customers informed about achievable advancements, driving innovation in new product development.

Does the environment affect your products?

Yes, we test our products in different environments to make sure they meet quality standards. We verify that they can operate at temperatures from -40 to 150 degrees Celsius, as required by Automotive Electronics Council Q100 Grade 0. Maintaining this level of quality over many years has always been a challenge, and Allegro has striven to do so.

What is your company doing to keep improving its products?

The most important thing we do is keep innovating. Our chief technology officer's office is constantly exploring what needs to evolve, improve, or be reimaged to stay ahead of the curve. While I may not have specific examples to share at the moment, our research and development efforts never stop. We are relentlessly focused on enhancing performance, minimising power consumption, and, as I said earlier, reducing product size because, as technology advances, the demand

for smaller, lighter, and more energy-efficient solutions continues to grow.

How do you help take care of the environment?

We prioritise automation and energy efficiency, which helps reduce environmental impact. We use less energy and operate more efficiently. Beyond improving our own energy use and recycling, we create products that help our customers to make their own designs more energy-efficient.

What is your role in the automation industry and electric vehicle market?

In markets like electric vehicles (EVs), clean energy, cloud computing, and automation, the focus is on energy efficiency and automation. Our products, including current sensors, gate driver ICs, and position sensors, are designed to enable energy conversion and precise control. For example, in factories, robotic arms rely on sensors and motor drivers to achieve accurate movement. Similarly, in cars, our sensors and drivers are used in power steering systems. Additionally, our power management ICs enable efficient, safe voltage conversion, a critical function in EVs and other energy-conscious applications. As industries push for smarter, more efficient systems, our technology helps drive the future of automation and energy optimisation.

Do you have plans to begin manufacturing in India?

Right now, we do not have any facilities or partners in India for manufacturing, but we have engineering and sales offices in India. This status may change as India continues to develop. As a fabless company, we design our products and have them manufactured by our fab partners in the United States, Taiwan, and Southeast Asia. We plan to start manufacturing in China soon. **EFY**